Box Office Smash or Bomb:
Forecasting Movie Revenue with Machine Learning and Text Embeddings

Executive summary · Erdős Fall 2026 Deep Learning Bootcamp

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64%	**59%**	**4.5×**	**×2.8**
of films rated “likely smash” broke even, vs 42% of all films	of films flagged “likely bomb” lost money, vs 28% of all films	gross-to-budget for “likely smash” films, vs 3.0× for all films	typical forecast miss, so use it to rank, not to budget

The bottom line

Film investment is a high-stakes, low-odds business: only about 43% of films earn the roughly 2.5 times their budget needed to break even, and the top 10% of films take 54% of all revenue. Our model forecasts a film's worldwide box office from information available before release. Tested on 674 held-out films with known budgets, it sorts projects into clearly different risk groups: films it rated as likely smashes broke even 64% of the time, against 42% for all films, while 59% of the films it flagged as likely bombs lost money outright. We recommend using it as a first-pass screen and an independent second opinion on a slate of projects, not as a precise revenue forecast.

Would it have picked better bets?

Model's call (forecast ÷ budget)	Share of films	Broke even (≥2.5×)	Lost money (<1×)	Total gross ÷ budget
Likely smash (≥ 2.5×)	25%	**64%**	12%	4.5×
Uncertain (1–2.5×)	57%	41%	25%	2.7×
Likely bomb (< 1×)	18%	18%	**59%**	1.7×
All test films	100%	42%	28%	3.0×

Back-test of the final XGBoost model on 674 held-out test films with a known budget (2025 dollars). Break-even rule of thumb: 2.5× budget in worldwide gross.

The model is most valuable at separating good bets from bad ones. As a group, the films it rated likely smashes returned 4.5 times their combined budget in worldwide gross, against 3.0 times for all films, and only 12% of them lost money. The films it flagged as likely bombs mostly lost money, and just 18% broke even. It also ranks well: 61% of the true top-10% earners were in the model's own top 10%. Budget is the strongest driver, followed by the production team's track record; notably, experienced producers, editors and composers predict success better than star power (an association, not proof of cause).

How stakeholders can use it

Decision	Who	Use the model?	How
Screening a slate of projects	Studios, greenlight committees	**Yes, as a first filter**	Prioritize “likely smash” projects; send “likely bomb” projects for closer review
Assembling the team	Producers	**Yes, directionally**	Compare scenarios, e.g. an experienced vs a first-time production team
Financing a film	Investors, lenders	**Yes, as a second opinion**	Benchmark the producer's projections; most precise above $66M
Release and marketing plans	Distributors, marketing	**Partly**	Rank upcoming titles to prioritize screens and promotional spend
Exact budgets, revenue targets or automatic rejections	All	**No**	Errors are too large, and it misses most eventual hits

The model’s practicality today

The model is ready to support screening and ranking, but it should not make go/no-go calls on its own. It is cautious: it labels only a quarter of films as likely smashes and catches about a third of the films (37%) that actually broke even, so a hard filter would reject many future hits, and 57% of films fall in an uncertain middle band where judgment must decide. A typical forecast is off by a factor of about 2.8 (a film predicted at $100M could plausibly earn $36M to $280M), though forecasts are tighter for budgets above $66M. It also relies on runtime and full crew details that are firm only late in production, and it was trained on films up to 2017, before streaming and the post-pandemic market.

Next steps: Report ranges with a break-even probability, retrain on recent releases including streaming revenue, and test on newer films before real use.